Secure DevOps

Setting up a secure environment for Ansible

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Today's Topics

- Many demos and tutorials skip over how to set up a secure environment for Ansible
  - SSH & GPG Keys, SSH-Agent, GPG-Agent
  - `pass` – The Unix Password Manager
  - Ansible Vault
- **SSH Setup Demo**
- **Ansible Setup Demo**
- **Ansible Demo**

SSH Setup Demo

Generate SSH Keys

- Generate an ED25519 SSH key with the comment “DemoSSHKey” and save it using that same filename – USE A PASSWORD to protect your SSH Key.

```
ssh-keygen -t ed25519 -C DemoSSHKey \
  -f /home/ansible_user/.ssh/DemoSSHKey
```

- This command breaks down as follows:
  - `-t ed25519`: Specifies the type of key to generate (ED25519).
  - `-C DemoSSHKey`: Adds the comment “DemoSSHKey” to the key
  - `-f .../DemoSSHKey`: Saves keys to `~/.ssh/DemoSSHKey` and `.pub`.

Upload SSH Public Key

- Copy your public SSH key and upload it to your server or cloud provider
- The public key is safe to share – that's the whole point
- For cloud providers (e.g., DigitalOcean), paste the key in the SSH Keys section of the console

```
cat ~/.ssh/DemoSSHKey.pub
```

RSA 4096 vs ED25519


```
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQD4kiiiQH2Avx4rzOsNf
fLvnEZAMuEF7CAOcpg3L6BO1aFu6jEFvr/NonnL1tPN/OwDoTeEMG4k7B
N3dAXFFkgFAGAT+2RNvTnFb+nhyReeEkwZg+Iw9+stfxITrtvKgoNQ76n
884P79FciMbn9QP6BmCXsTUQhMFMvWHPO3rwYXFt9f/pUgW6QCmpIw5i3f
KgQfOWGFVsjPWXMCXrNiIDDk/7iVUqXg8FKFUvQUG1DGDcspPdelOgsJSX
y14Wov4l4sSlViYqe6WQXKA6aIvsQGk+DfZErKPWMg+2OLCAYYQeD1+r4Q
iLbMHbHNK26WBmkjzvOpgkbEdBdrscRaTi8AFCWelyqsoxxz4GEKOCG7NoB
idlOKTf95OqDpmAnZcwRDTLWbYlLBxgL8+Pmjgku9Pnzhf2qipQs94A+te
EgZiPlpIXSitqcBmJOaamCTMrYE9BkylM7hvGkfx7QZYWktQkgdPKKl8ac
KDM77nc1SWc1vQ2pHhq53xk08cNWrOMZaxqZaDMGncxp83GnOEjkZhr6Xw
9T5/ Demo_SSH_RSA_Key
```



```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDGwqTc2eyAVxspuLyNERj
9ktEHdkRwdvKpAaO8ZP68 Demo_SSH_ED25519_Key
```

Configure SSH Client

```
Host ansible
    HostName 192.168.122.100
    User root
    IdentityFile ~/.ssh/DemoSSHKey
```

`Host ansible`: Creates an alias “ansible” for easy connection.

`HostName 192.168.122.100`: Sets the server’s IP address.

`User root`: Uses “root” as the default username.

`IdentityFile ~/.ssh/DemoSSHKey`: Specifies the SSH key for secure login.

SSH to ansible, part 1

```
$ ssh ansible
Enter passphrase for key '~/.ssh/DemoSSHKey':
Welcome to Ubuntu 24.04 LTS

 System load: 0.08 Processes: 104
 Usage of /: 19.2% Users logged in: 0
 Memory usage: 36% IPv4: 192.168.122.100

root@ansible:~#
```

- Notice that I have to enter a password to use my SSH key.
- Let's fix that by using SSH agent
- Add this line to your config file:

  `AddKeysToAgent yes`

```
Host ansible
    HostName 192.168.122.100
    User root
    AddKeysToAgent yes
    IdentityFile ~/.ssh/DemoSSHKey
```

CyberSecurity Time

- Update the `sshd_config` on the ansible server
  - Disable root login: `PermitRootLogin no` ensures that root cannot access the server via SSH.
  - Enforce public key authentication: `PubkeyAuthentication yes` and `PasswordAuthentication no` require that all users authenticate with an SSH key.
  - Do not forget to restart SSHD: `systemctl restart ssh.service`

```
# Authentication:

#LoginGraceTime 2m
PermitRootLogin no <--
#StrictModes yes
#MaxAuthTries 6

PubkeyAuthentication yes <--

...

PasswordAuthentication no <--
#PermitEmptyPasswords no
```

Update ssh config

- Update my ssh client config to use `ansible_user` instead of `root`

```
Host ansible
    HostName 192.168.122.100
    User ansible_user
    AddKeysToAgent yes
    IdentityFile ~/.ssh/
        DemoSSHKey
```

```
$ ssh ansible
root@192.168.122.100: Permission denied
  (publickey).

$ vim ~/.ssh/config

$ ssh ansible
Welcome to Ubuntu 24.04 LTS
...
ansible_user@ansible:~$
```

SSH Setup Summary

- Created SSH keys, uploaded the public key to cloud provider
- Configured our client to use SSH Agent with SSH keys
- Connected to our ansible server, using SSH keys
- Hardened SSHd on the ansible server
- Verified that our settings worked
- **QUESTION: Are we doing DevOps?**

Ansible Setup Demo

Install Ansible

```
ansible_user@ansible:~$ ./install_ansible.sh
Reading package lists... Done
Building dependency tree... Done
...
The following NEW packages will be installed:
  python3 python3-pip python3-venv ...
...
Do you want to continue? [Y/n]
```

- Use `git` to download the install script
- `chmod +x install_ansible.sh`
- `./install_ansible.sh`

Logout & Login

- After install, `~/.local/bin` is added to `PATH`
- But you need to **logout and log back in** for the change to take effect

Ansible Installed

```
ansible_user@ansible:~$ ansible --version
ansible [core 2.18.1]
  config file = None
  configured module search path =
    ['/home/ansible_user/.ansible/plugins/modules',
     '/usr/share/ansible/plugins/modules']
  ansible python module location =
    /home/ansible_user/.local/share/uv/...
  executable location =
    /home/ansible_user/.local/bin/ansible
  python version = 3.12.3
  jinja version = 3.1.4
  libyaml = True
```

Ansible Demo

Our First Ansible Playbook

- Create an inventory file
- Write a playbook
- Run the playbook

$ cat inventory.ini

```
[local]
localhost ansible_connection=local
```

$ cat ping.yml

```
---
- name: Ping localhost
  hosts: local
  tasks:
    - name: Ping the localhost
      ping:
```

```
ansible_user@ansible:~/secure_devops_demo$ ansible-playbook -i inventory.ini ping.yml

PLAY [Ping localhost] ***************************************************
TASK [Gathering Facts] ***************************************************
ok: [localhost]
TASK [Ping the localhost] ************************************************
ok: [localhost]
PLAY RECAP ***************************************************************
localhost : ok=2 changed=0 unreachable=0 failed=0
```

ansible.cfg

```
[defaults]
inventory = inventory.ini
command_warnings = False
deprecation_warnings = False
interpreter_python =
    /usr/bin/python3
```

Add the inventory location and turn off those warnings

Now you don't have to pass the `-i inventory.ini` each time you run a playbook

```
ansible_user@ansible:~/secure_devops_demo$
  ansible-playbook ping.yml

PLAY [Ping localhost] *****************
TASK [Gathering Facts] ****************
ok: [localhost]
TASK [Ping the localhost] *************
ok: [localhost]
PLAY RECAP ****************************
localhost : ok=2 changed=0 failed=0
```

Update ansible with Ansible

```yaml
---
- name: Update and upgrade packages
    on localhost
  hosts: local
  become: yes  tasks:
    - name: Update apt cache
      apt:
        update_cache: yes
    - name: Upgrade all packages
      apt:
        upgrade: dist
```

- Notice that we are using `become` to use sudo
- It failed — we need a password to use sudo
- We need a secure way to store and provide that password

```
ansible_user@ansible:~/secure_devops_demo$ ansible-playbook update.yml
[WARNING]: Error getting vault password file (default):
  The vault password file /nonexistent was not found
[ERROR]: The vault password file /nonexistent was not found
```

Ansible with sudo

```
ansible_user@ansible:~/secure_devops_demo$
  ansible-playbook update.yml --ask-become-pass
BECOME password:

PLAY [Update and upgrade packages on localhost] ********
TASK [Gathering Facts] *********************************
ok: [localhost]
TASK [Update apt cache] ********************************
changed: [localhost]
```

- It worked!
- Typing passwords each time you run a playbook is no good.
- Let's fix that.

Storing Secrets in Ansible

- Ansible can store secrets in a “vault”
  - Uses AES 256 to encrypt the data
  - Needs to be unlocked before using
- You could store the `ansible_user` password for sudo in the vault and run the playbook:

  ```
  ansible-playbook update.yml -ask-vault-pass
  ```
- **Not much better than just running:**

  ```
  ansible-playbook update.yml -ask-become-pass
  ```

Ansible Vault

```
ansible_user@ansible:~/secure_devops_demo$ ansible-vault view vault.yml
Vault password:
ansible_become_password: ************
```

```
ansible_user@ansible:~/secure_devops_demo$ cat vault.yml
$ANSIBLE_VAULT;1.1;AES256
6136616539613930613666366663430656266383433333662643930636431 30
3366366336363336363432336131343539383632313330393334363133383 0
3630653331363339313634343631363433353633353333333636439393636 36
3866396230363336393930613338363133383633373833333036363337363 1
```

Let's use pass

```
ansible_user@ansible:~/secure_devops_demo$ ./install_setup_pass.sh
Installing pass, the Unix password manager...
[sudo] password for ansible_user:

gpg: directory '/home/ansible_user/.gnupg' created
gpg: keybox '/home/ansible_user/.gnupg/pubring.kbx' created
Generating a new GPG key for pass...
...
Password store initialized for 2466552D8747919
Pass initialized with GPG key ID: 2466552D8747919
Enter the password for the Ansible Vault:

Enter contents of ansible/vault_password and press Ctrl+D when finished:

Ansible Vault password has been securely stored in pass under
  'ansible/vault_password'.
ansible_user@ansible:~/secure_devops_demo$ pass ansible/vault_password
```

pass uses GPG

pass uses GPG to encrypt/decrypt passwords

```
ansible_user@ansible:~$ tree
  ~/.password-store/
/home/ansible_user/.password-store/
  ansible
    vault_password.gpg
```

pass just stores each entry as its own GPG encrypted file

GPG uses an agent, just like the SSH Agent

```
/run/user/1000/gnupg/S.gpg-agent

SSH Agent PID:
25296 ssh-agent
```

```
ansible_user@ansible:~$ file
  ~/.password-store/ansible/
    vault_password.gpg
...PGP RSA encrypted session key
  - keyid: DEF15331 A3CB758D
  RSA (Encrypt or Sign) 2048b

ansible_user@ansible:~$ xxd
  ~/.password-store/ansible/
    vault_password.gpg
00000000: 8501 0c03 def1 5331
00000010: 8ee9 9d32 2e5a 0c68
00000020: bf34 3e73 2e26 32b6
...
```

Update `ansible.cfg` for `pass`

- Update the ansible.cfg to call a script to use pass to unlock our vault

```
[defaults]
inventory = inventory.ini
command_warnings = False
deprecation_warnings = False
interpreter_python = /usr/bin/python3
vault_password_file =
    /home/ansible_user/bin/get_vault_pass.sh
```

```
#!/bin/bash
pass
↪  ansible/vault_password
```

- Wait, what were we doing again?
- Updating the packages on the ansible server, securely using NO PLAINTEXT passwords.

Update our `update.yml`

```yaml
---
- name: Update and upgrade packages
    on localhost
  hosts: local
  become: yes
  vars_files:    - vault.yml  tasks:
    - name: Update apt cache
      apt:
        update_cache: yes
    - name: Upgrade all packages
      apt:
        upgrade: dist
```

```
ansible_user@ansible:~/secure_devops_demo$
  ansible-playbook update.yml

PLAY [Update and upgrade packages
  on localhost] ********************

TASK [Gathering Facts] *************
ok: [localhost]

TASK [Update apt cache] ************
changed: [localhost]

TASK [Upgrade all packages] ********
ok: [localhost]

PLAY RECAP *************************
localhost : ok=3 changed=1
  unreachable=0 failed=0
```

Okay...

- Couldn't I have just run:

  ```
  apt update && apt upgrade -y
  ```

- **Yes, but that's not DevOps**
  - It doesn't scale to dozens / hundreds / thousands of machines
  - Error prone
  - You don't get "infrastructure as code"

Let’s spin up more machines

- 3 local VMs on QEMU/KVM (no internet needed) — cloud-init injects your public key automatically

BEFORE: `inventory.ini`

```
[servers]
```

```
$ bin/create_target_vms.sh
$ bin/update_local_inventory.sh
```

AFTER: `inventory.ini`

```
[servers]
target-1 ansible_host=192.168.122.101
target-2 ansible_host=192.168.122.102
target-3 ansible_host=192.168.122.103
```

Ansible to update `.bashrc`

```yaml
---
- hosts: localhost
  tasks:
    - name: Ensure ssh-agent is started
        and environment variables are saved
      blockinfile:
        path: ~/.bashrc
        marker: "# {mark} Ansible SSH
          agent configuration"
        block: |
          # Ansible modified: SSH agent setup
          if [ -z "$SSH_AUTH_SOCK" ]; then
            eval "$(ssh-agent -s)" >
              ~/.ssh-agent-variables
            ssh-add ~/.ssh/DemoSSHKey
          fi
          if [ -f ~/.ssh-agent-variables ]
          then
            source ~/.ssh-agent-variables
          fi
        create: yes
```

Now the `.bashrc` file is updated to start the SSH-Agent when the user logs in

```bash
# BEGIN Ansible SSH agent
#   configuration
# Ansible modified: SSH agent
#   setup
if [ -z "$SSH_AUTH_SOCK" ]; then
    eval "$(ssh-agent -s)" >
      ~/.ssh-agent-variables
    ssh-add ~/.ssh/DemoSSHKey
fi
if [ -f ~/.ssh-agent-variables ]
then
    source ~/.ssh-agent-variables
fi
# END Ansible SSH agent
#   configuration
```

Idempotent

```
ansible_user@ansible:~/secure_devops_demo$
  ansible-playbook add_ssh_key.yml

PLAY [localhost] ***********************
TASK [Gathering Facts] *****************
ok: [localhost]
TASK [Ensure ssh-agent is started...] **
changed: [localhost]

PLAY RECAP *****************************
localhost : ok=2 changed=1

ansible_user@ansible:~/secure_devops_demo$
  ansible-playbook add_ssh_key.yml

PLAY [localhost] ***********************
TASK [Gathering Facts] *****************
ok: [localhost]
TASK [Ensure ssh-agent is started...] **
ok: [localhost]

PLAY RECAP *****************************
localhost : ok=2 changed=0
```

- Idempotency ensures repeatability: Running the same operation multiple times produces the same outcome without unintended side effects.
- Prevents duplication and redundancy: Ensures that changes are only applied if necessary, making configuration management predictable and reliable.

Now we are ready

- Oh, right. First time SSHing, getting the fingerprint message.
- Also, we are trying to connect as `ansible_user`

```
ansible_user@ansible:~/secure_devops_demo$ ansible-playbook ping-servers.yml

PLAY [Ping servers] ****************************************************
TASK [Gathering Facts] *************************************************
fatal: [192.168.122.101]: UNREACHABLE! => {"changed": false, "msg": "Failed
  to connect to the host via ssh: ansible_user@192.168.122.101: Permission
  denied (publickey).", "unreachable": true}
The authenticity of host '192.168.122.102 (192.168.122.102)' can't be
  established.
ED25519 key fingerprint is SHA256:PzXwrkP3W+...
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
fatal: [192.168.122.102]: UNREACHABLE! => {"changed": false, "msg": "Failed
  to connect to the host via ssh: Permission denied (publickey)."}
```

Add `ansible_user` to servers

```yaml
---
- name: Add ansible_user to the sudo
    group, set up SSH key, and set
    password
  hosts: servers
  become: true
  remote_user: root  vars_files:
    - vault.yml
  tasks:
    - name: Install passlib on remote
        hosts (if required)
      ansible.builtin.package:
        name: python3-passlib
        state: present
    - name: Ensure ansible_user is in
        the sudo group with a hashed
        password
      user:
        name: ansible_user
        groups: sudo
        append: true
        password: "{{ ansible_user_password
          | password_hash('sha512') }}"
```

```
ansible_user@ansible:~/secure_devops_demo$
  ansible-playbook add_ansible_user.yml

PLAY [Add ansible_user to the sudo
  group, set up SSH key, and set
  password] ************************

TASK [Gathering Facts] *************
ok: [192.168.122.101]
ok: [192.168.122.103]
ok: [192.168.122.102]

TASK [Install passlib...] **********
ok: [192.168.122.101]
ok: [192.168.122.102]
ok: [192.168.122.103]

TASK [Ensure ansible_user...] ******
changed: [192.168.122.101]
changed: [192.168.122.102]
changed: [192.168.122.103]

TASK [Create .ssh directory...] ****
ok: [192.168.122.101]
ok: [192.168.122.102]
ok: [192.168.122.103]
```

Switch SSH config to `ansible_user`

```
ansible_user@ansible:~/secure_devops_demo$ cat ~/.ssh/config
# BEGIN cyberforge-demo
Host server1
    HostName 192.168.122.101
    User ansible_user
    IdentityFile ~/.ssh/DemoSSHKey
    AddKeysToAgent yes
    StrictHostKeyChecking accept-new

Host server2
    HostName 192.168.122.102
    User ansible_user
    IdentityFile ~/.ssh/DemoSSHKey
    AddKeysToAgent yes
    StrictHostKeyChecking accept-new

Host server3
    HostName 192.168.122.103
    User ansible_user
    IdentityFile ~/.ssh/DemoSSHKey
    AddKeysToAgent yes
    StrictHostKeyChecking accept-new
# END cyberforge-demo
```

Verify `ansible_user` can connect

```
ansible_user@ansible:~/secure_devops_demo$ ansible-playbook ping-servers.yml

PLAY [Ping servers] ***************************************************

TASK [Gathering Facts] ************************************************
ok: [192.168.122.101]
ok: [192.168.122.102]
ok: [192.168.122.103]

TASK [Ping the localhost] *********************************************
ok: [192.168.122.101]
ok: [192.168.122.102]
ok: [192.168.122.103]

PLAY RECAP ************************************************************
192.168.122.101 : ok=2 changed=0 unreachable=0 failed=0
192.168.122.102 : ok=2 changed=0 unreachable=0 failed=0
192.168.122.103 : ok=2 changed=0 unreachable=0 failed=0
```

SSHD Hardening

```yaml
---
- name: Configure SSHD for Ubuntu
    22.04, 24.04, and 24.10
  hosts: servers
  become: true
  vars_files:    - vault.yml  tasks:
    - name: Ensure root login is
        disabled
      lineinfile:
        path: /etc/ssh/sshd_config
        regexp: '^#?PermitRootLogin'
        line: 'PermitRootLogin no'
        state: present
    - name: Ensure public key
        authentication is enabled
      lineinfile:
        path: /etc/ssh/sshd_config
        regexp: '^#?PubkeyAuthentication'
        line: 'PubkeyAuthentication yes'
        state: present
    - name: Disable password
        authentication
      lineinfile:
        path: /etc/ssh/sshd_config
        regexp: '^#?PasswordAuthentication'
        line: 'PasswordAuthentication no'
        state: present
    - name: Restart SSHD
      service:
        name: ssh
        state: restarted
```

```
ansible_user@ansible:~/secure_devops_demo$
  ansible-playbook
    sshd_hardening-servers.yml

PLAY [Configure SSHD for Ubuntu
  22.04, 24.04, and 24.10] *********

TASK [Gathering Facts] *************
ok: [192.168.122.101]
ok: [192.168.122.103]
ok: [192.168.122.102]

TASK [Ensure root login is
  disabled] **************************
changed: [192.168.122.101]
changed: [192.168.122.102]
changed: [192.168.122.103]

TASK [Ensure public key auth...] ***
changed: [192.168.122.101]
changed: [192.168.122.102]
changed: [192.168.122.103]

TASK [Disable password auth...] ****
changed: [192.168.122.101]
changed: [192.168.122.102]
```

Now we can ping servers

```
ansible_user@ansible:~/secure_devops_demo$ ansible-playbook ping-servers.yml

PLAY [Ping servers] ************************************************

TASK [Gathering Facts] *********************************************
ok: [192.168.122.101]
ok: [192.168.122.102]
ok: [192.168.122.103]

TASK [Ping the localhost] ******************************************
ok: [192.168.122.101]
ok: [192.168.122.102]
ok: [192.168.122.103]

PLAY RECAP *********************************************************
192.168.122.101 : ok=2 changed=0 unreachable=0 failed=0
192.168.122.102 : ok=2 changed=0 unreachable=0 failed=0
192.168.122.103 : ok=2 changed=0 unreachable=0 failed=0
```

Ad-hoc commands

```
ansible servers -m shell -a
"df -h"
```

```
192.168.122.101 | CHANGED | rc=0 >>
Filesystem Size Used Avail Use% Mounted
/dev/vda1 9.6G 1.7G 7.9G 18% /
tmpfs 46M 1008K 45M 3% /run
...
192.168.122.102 | CHANGED | rc=0 >>
Filesystem Size Used Avail Use% Mounted
/dev/vda1 9.6G 1.7G 7.9G 18% /
...
192.168.122.103 | CHANGED | rc=0 >>
Filesystem Size Used Avail Use% Mounted
/dev/vda1 9.6G 1.7G 7.9G 18% /
```

```
ansible servers -a "ss -tuln"
```

```
192.168.122.101 | CHANGED | rc=0 >>
Netid State Recv-Q Send-Q
  Local Address:Port
udp UNCONN 0 0
  127.0.0.53%lo:53
tcp LISTEN 0 4096
  127.0.0.53%lo:53
tcp LISTEN 0 128
  0.0.0.0:22
tcp LISTEN 0 128
  [::]:22
192.168.122.102 | CHANGED | rc=0 >>
...
192.168.122.103 | CHANGED | rc=0 >>
...
```

Summary

- DevOps is awesome!
- Many demos and tutorials skip over how to set up a secure environment for Ansible
  - SSH & GPG Keys
  - SSH-Agent
  - GPG-Agent
  - `pass` – The Unix Password Manager
  - Ansible Vault

Questions?

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